

8. QUEUE USING ARRAY

SOURCE CODE:

```
#include <stdio.h>
#include <stdlib.h>

#define MAX_SIZE 100

int queue[MAX_SIZE];
int front = -1;
int rear = -1;

void enqueue(int item) {
    if (rear == MAX_SIZE - 1) {
        printf("Queue Overflow\n");
        return;
    }
    if (front == -1 && rear == -1) {
        front = 0;
        rear = 0;
    } else {
        rear++;
    }
    queue[rear] = item;
    printf("%d enqueued to queue\n", item);
}

int dequeue() {
    int item;
    if (front == -1 || front > rear) {
        printf("Queue Underflow\n");
        return -1;
    }
    item = queue[front];
    front++;

    // Reset queue if it's empty now
    if (front > rear) {
        front = -1;
        rear = -1;
    }
    return item;
}

void display() {
    if (front == -1 || front > rear) {
        printf("Queue is empty\n");
        return;
    }
    printf("Queue elements: ");
    for (int i = front; i <= rear; i++) {
        printf("%d ", queue[i]);
    }
    printf("\n");
}

int main() {
```

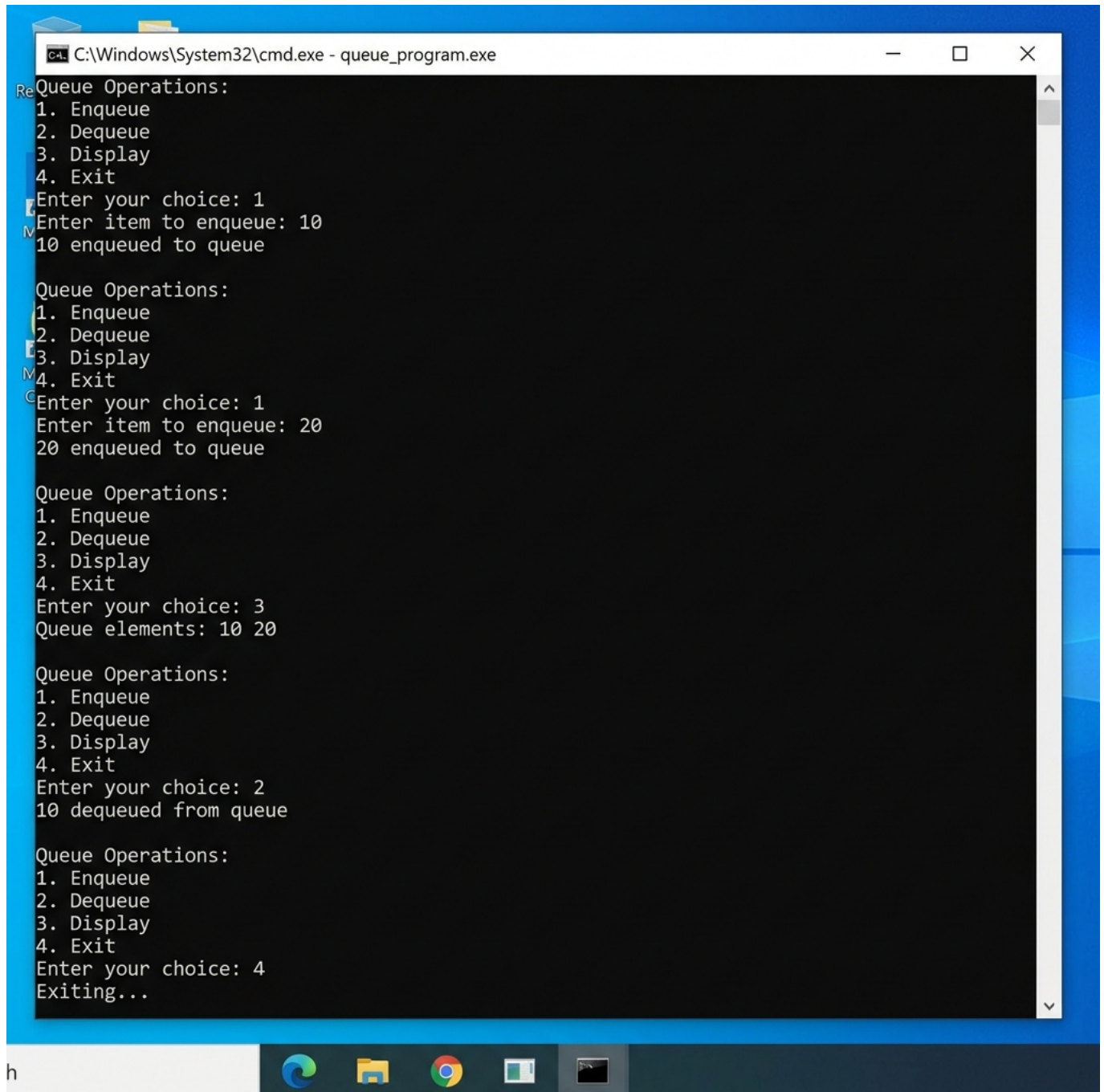
```

int choice, item;
while (1) {
    printf("\nQueue Operations:\n");
    printf("1. Enqueue\n");
    printf("2. Dequeue\n");
    printf("3. Display\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            printf("Enter item to enqueue: ");
            scanf("%d", &item);
            enqueue(item);
            break;
        case 2:
            item = dequeue();
            if (item != -1) {
                printf("%d dequeued from queue\n", item);
            }
            break;
        case 3:
            display();
            break;
        case 4:
            exit(0);
        default:
            printf("Invalid choice\n");
    }
}
return 0;
}

```

PROGRAM OUTPUT



The screenshot shows a Windows desktop with a blue background. A command prompt window is open, displaying the output of a program titled 'queue_program.exe'. The window title bar reads 'C:\Windows\System32\cmd.exe - queue_program.exe'. The program's output is as follows:

```
Queue Operations:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter item to enqueue: 10
10 enqueued to queue

Queue Operations:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter item to enqueue: 20
20 enqueued to queue

Queue Operations:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue elements: 10 20

Queue Operations:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2
10 dequeued from queue

Queue Operations:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 4
Exiting...
```

The taskbar at the bottom shows the Start button and several application icons: Edge, File Explorer, Chrome, and a task view icon.